

Ren Wang

✉ renwang@cmlab.csie.ntu.edu.tw | 🌐 <https://renwang0508.github.io>

Education

National Taiwan University , Taipei, Taiwan <i>Ph.D. Candidate in Computer Science and Information Engineering</i> Advisor: Yung-Yu Chuang	2021–present
National Tsing Hua University , Hsinchu, Taiwan <i>M.S. in Computer Science</i> Advisor: Hwann-Tzong Chen	2013–2015
National Tsing Hua University , Hsinchu, Taiwan <i>B.S. in Computer Science</i>	2009–2013

Work Experience

Google Taiwan , New Taipei City, Taiwan <i>Senior Software Engineer, Pixel Camera</i>	10/2024–present
MediaTek Inc. , Hsinchu, Taiwan <i>Senior Engineer, Multimedia Development Div. XVI</i>	03/2023–06/2024
<i>Senior Engineer, Multimedia Technology Development Div.</i>	06/2020–02/2023
<i>Engineer, Multimedia Technology Development Div.</i>	02/2018–05/2020
<i>Engineer, Advanced Communication Technology Div.</i>	12/2015–01/2018

Selected Projects

Instant Night Sight , Pixel Camera, Google 🔗	2025–2026
---	-----------

ACHIEVEMENTS

- Co-developed the core model for ultra-short exposures in low light, deployed since the Pixel 11 series.

RESPONSIBILITIES

- Led model optimization strategies and development via quantization and distillation techniques.
- Collaborated closely with cross-functional teams, including research, silicon, and tooling.

Pro Zoom , Pixel Camera, Google 🔗	2024–2025
--	-----------

ACHIEVEMENTS

- Co-developed the core generative model for up to 120x zoom, deployed since the Pixel 11 series.

RESPONSIBILITIES

- Optimized single-step diffusion models without quality degradation via quantization techniques.
- Investigated the impact of design space on sampling quality of advanced generative frameworks, including diffusion, flow matching, and EDM.

Unified Codebase for AI-ISP Algorithm Development , MediaTek	2023–2024
---	-----------

ACHIEVEMENTS

- Integrated 4 distinct AI-ISP projects, including image/video denoising, enhancement, and remosaicing.
- Enabled 10+ engineers to accelerate algorithm development by leveraging the codebase.

RESPONSIBILITIES

- Led a team of 5+ engineers to build the codebase infrastructure and establish collaboration guidelines.
- Optimized the software architecture to support the diverse requirements of AI-ISP projects.

AI-Noise Reduction in Low-Light Captures, MediaTek

2017–2022

ACHIEVEMENTS

- Contributed to scaling this project across 25 smartphone series from June 2019 to Feb 2022, where the clients included OPPO, vivo, Xiaomi, realme, and OnePlus.

RESPONSIBILITIES

- Applied cutting-edge AI techniques to image denoising on MediaTek’s image signal processors.
- Optimized neural networks to reduce the computation time and memory/power consumption on MediaTek’s deep learning accelerators.
- Collaborated closely with relevant projects, such as image enhancement, demosaicing, and HDR.

Activity and Gesture Recognition, MediaTek

2016–2017

ACHIEVEMENTS

- Contributed to deploying this project on Helio X30 in 2017, where the client was Meizhu.

RESPONSIBILITIES

- Implemented statistical models and optimized their computational costs on mobile platforms, where the models included SVM, GMM, and HMM.
- Prototyped deep learning models for gesture recognition using IMU sensors.

Publications

- **Ren Wang** and Yung-Yu Chuang, “Uncertainty-guided latent diffusion models for faithful super resolution,” in *Proceedings of the IEEE International Conference on Image Processing (ICIP)*, Sept 2026.
- Shao-Hao Lu, **Ren Wang**, Ching-Chun Huang, and Wei-Chen Chiu, “Boosting diffusion guidance via learning degradation-aware models for blind super resolution,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Feb 2025. **[Oral Presentation]**
- **Ren Wang**, Yu-Lun Liu, Yu-Hao Huang, and Ning-Hsu Wang, “Methods and apparatuses of depth estimation from focus information,” U.S. Patent 11,967,096, issued Apr 23, 2024.
- Ning-Hsu Wang, **Ren Wang**, Yu-Lun Liu, Yu-Hao Huang, Yu-Lin Chang, Chia-Ping Chen, and Kevin Jou, “Bridging unsupervised and supervised depth from focus via all-in-focus supervision,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, Oct 2021.
- Chien-Chuan Su, **Ren Wang**, Hung-Jin Lin, Yu-Lun Liu, Chia-Ping Chen, Yu-Lin Chang, and Soo-Chang Pei, “Explorable tone mapping operators,” in *Proceedings of the International Conference on Pattern Recognition (ICPR)*, Jan 2021.
- Ke-Chi Chang, **Ren Wang**, Hung-Jin Lin, Yu-Lun Liu, Chia-Ping Chen, Yu-Lin Chang, and Hwann-Tzong Chen, “Learning camera-aware noise models,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, Aug 2020.

Honors

- | | |
|---|------------|
| • Google Gold Perfy Award (for Instant Night Sight) | Aug 2026 |
| • Google Spot Bonus (2 times; for recognized achievements) | 2025, 2026 |
| • MediaTek vAward (16 times; for exemplary performance) | 2015–2024 |
| • MediaTek Project Award (for Dimensity 1000) | July 2020 |
| • Valedictorian , NTHU CS | June 2013 |
| • Finalist of the Senior Project Contest , NTHU CS | Oct 2012 |

Academic Activities

Paper Reviewer

- IEEE Transactions on Image Processing

- Conference on Neural Information Processing Systems (NeurIPS) 2026
- European Conference on Computer Vision (ECCV) 2026
- IEEE International Conference on Image Processing (ICIP) 2026
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2025

Teaching Assistant, National Tsing Hua University

- ISA 525700: Computer Vision for Visual Effects Spring 2015
- EECS 111000: Introduction to Programming (C language) Fall 2013, 2014
- CS 321100: Introduction to Communication Fall 2014

Technical Skills

Expertise Generative AI, Deep Learning, Computational Photography, Low-Level Vision

Programming Python, C/C++, Git, PyTorch, JAX (Flax/Optax), TensorFlow, scikit-learn, OpenCV